

# Electronic circuits

## Exam 2019

Prepared by :

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Department of Electronics and Communication  
Even Semester Examinations .Academic Year 2018/2019  
Subject: Electronics Circuit

Dr. A. Bawazir  
Date: 8/7/2019

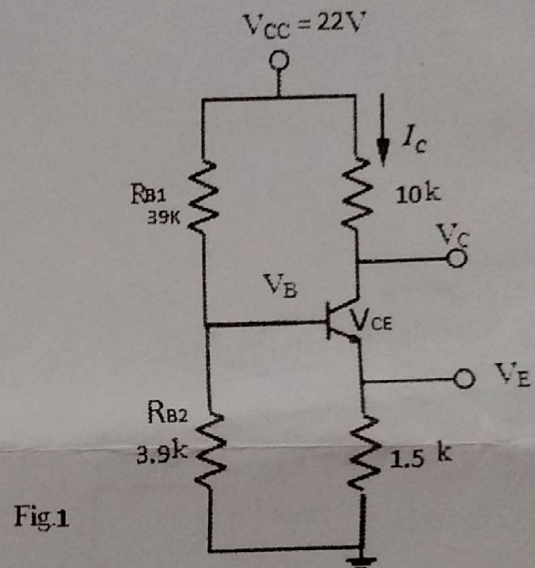
Class: B2CS/E  
ANSWER ANY FOUR QUESTIONS

Time: 2 Hours

**Question 1**  
Calculate  $V_{CE}$  and  $I_C$  of Fig.1

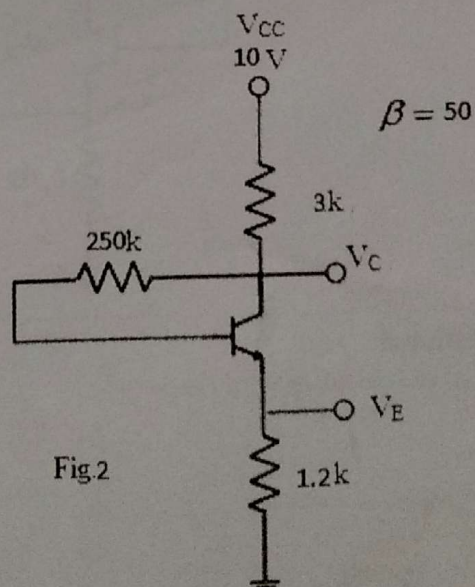
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~~$I_E = 1.02$~~   
 ~~$I_C = 0.998$~~   
 ~~$V_{CE} = 12.63$~~



**Question 2**  
Calculate  $I_E$  and  $V_{CE}$  using voltage feedback

$I_E = 1.02$   
 $V_{CE} = 5.716$



### Question 3

Calculate  $V_{CE}$  and  $I_C$  of Fig.3

$$\begin{aligned} I_B &= 36.34 \mu A \\ I_C &= 3.634 \text{ mA} \\ V_{CE} &= 9.098 \text{ V} \end{aligned}$$

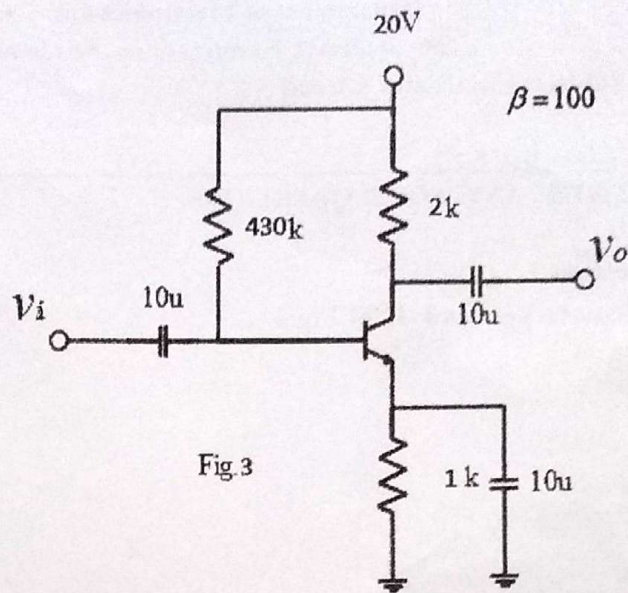


Fig.3

### Question 4

Calculate  $\beta$ ,  $V_{CC}$ ,  $I_C$ ,  $R_B$

$$\begin{aligned} \beta &= 154 \\ V_{CC} &= 17.71 \text{ V} \\ I_C &= I_E = 7.08 \text{ mA} \\ R_B &= 745.7 \Omega \end{aligned}$$

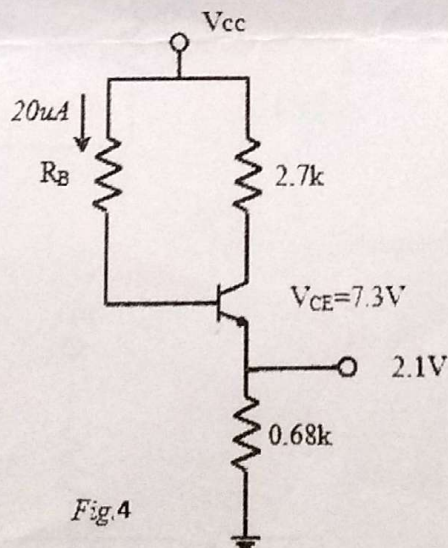


Fig.4

### Question 5

Sketch and calculate  $V_o$  of Fig.5

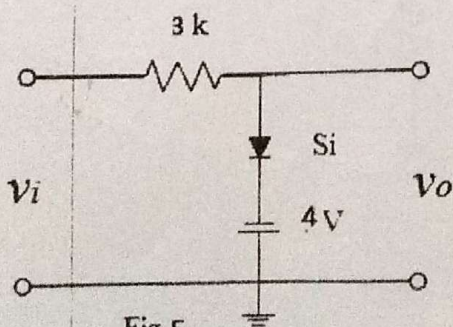
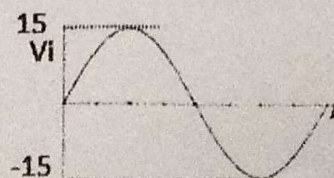


Fig.5

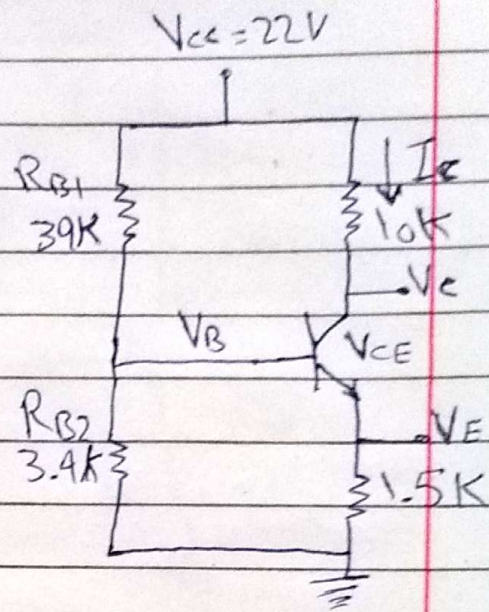
$$+ \frac{1}{2}) V_o = -4 + 0.7 = -3.3 \text{ V}$$



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Q.1)  $V_{CE} = ?$ ,  $I_C = ?$



$$V_B = \frac{R_{B2} \cdot V_{CC}}{R_{B1} + R_{B2}} = \frac{3.4 \times 22}{39 + 3.4} = 2 \text{ V}$$

$$V_{BE} = V_B - V_E \Rightarrow V_E = V_B - V_{BE} = 2 - 0.7 = 1.3 \text{ V}$$

$$I_C \approx I_E = \frac{V_E}{R_E} = \frac{1.3}{1.5} = 0.866 \text{ mA}$$

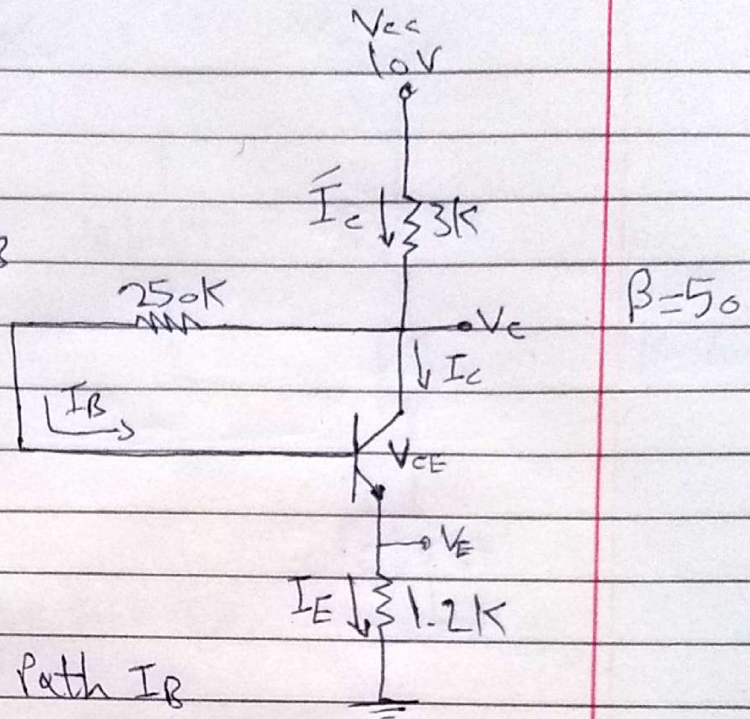
$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

$$V_{CE} = 22 - 0.866(10 + 1.5) = 12.041 \text{ V}$$

Q.2)  $I_E = ?$ ,  $V_{CE} = ?$

$$\bar{I}_C = I_C + I_B \approx I_C = \beta I_B$$



Apply KVL on the path  $I_B$

$$V_{CC} - \bar{I}_C R_C - I_B R_B - V_{BE} - I_E R_E = 0$$

$$V_{CC} - \beta I_B R_C - I_B R_B - V_{BE} - (\beta + 1) I_B R_E = 0$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_C + R_B + (\beta + 1) R_E} = \frac{10 - 0.7}{50 \times 3 + 250 + (50 + 1) \times 1.2}$$

$$I_B = 20.16 \mu A$$

$$I_C = \beta I_B = 50 \times 20.16 = 1.008 \text{ mA}$$

$$I_E = I_C + I_B = 1.008 + 0.02016 = 1.028 \text{ mA} \approx I_C$$

Apply KVL on the path  $V_{CE}$

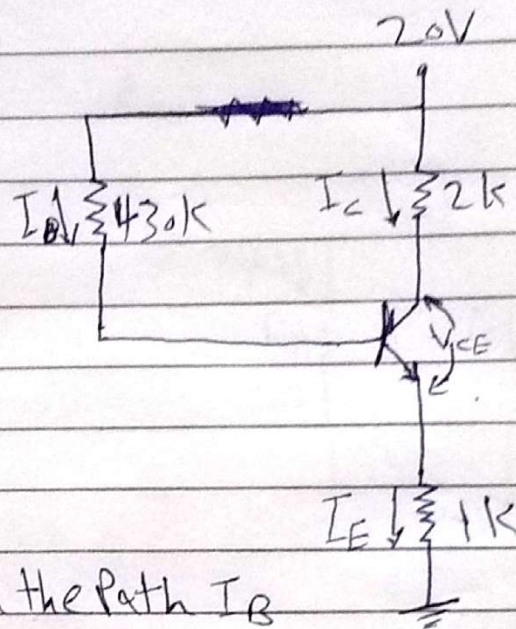
$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$(I_E \approx I_C)$

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

$$V_{CE} = 10 - 1.008 (3 + 1.2) = 5.76 \text{ V}$$

Q.3)  $V_{CE} = ?$ ,  $I_C = ?$



$\beta = 100$

APPLY KVL act on the Path  $I_B$

$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

$$I_E = I_C + I_B = \beta I_B + I_B = (\beta + 1) I_B$$

$$V_{CC} - I_B R_B - V_{BE} - (\beta + 1) I_B R_E = 0$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1) R_E} = \frac{20 - 0.7}{430 + (100 + 1) \times 1} = 36.34 \mu A$$

$$I_C = \beta I_B = 100 \times 36.34 = 3.634 \text{ mA}$$

APPLY KVL act on the Path  $V_{CE}$

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$I_E \approx I_C$$

$$V_{CC} - I_C R_C - V_{CE} - I_C R_E = 0$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E) = 20 - 3.634 (2 + 1) = 9.098 \text{ V}$$

Q.4)  $\beta = ?$ ,  $V_{CC} = ?$ ,  $I_C = ?$ ,  $R_B = ?$

$$I_C = \beta I_B \Rightarrow \beta = \frac{I_C}{I_B}$$

$$I_C \approx I_E = \frac{V_{RE}}{R_E}$$

$$I_E = \frac{2.1}{0.68} = 3.088 \text{ mA} \approx I_C$$

$$\beta = \frac{I_C}{I_B} = \frac{3.088}{20 \times 10^{-3}} = 154$$

APPLY KVL act on the Path  $V_{CE}$

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$I_C \approx I_E$$

$$V_{CC} = V_{CE} + I_E (R_C + R_E) = 7.3 + 3.08(2.7 + 0.68)$$

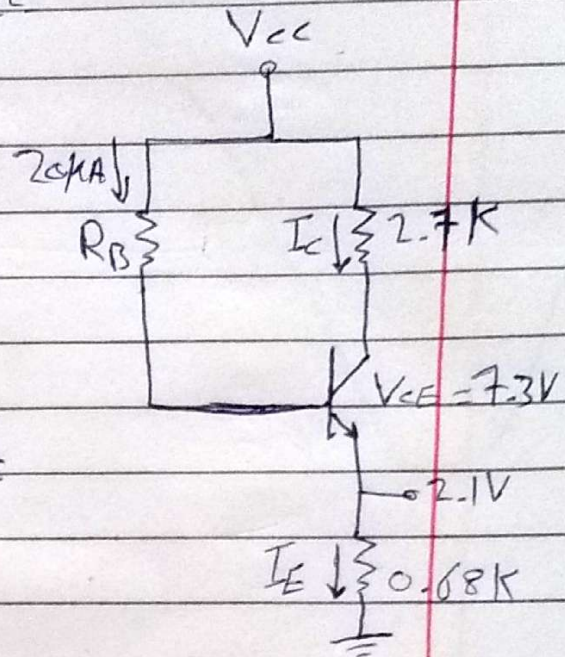
$$V_{CC} = 17.71 \text{ V}$$

APPLY KVL act on the Path  $I_B$

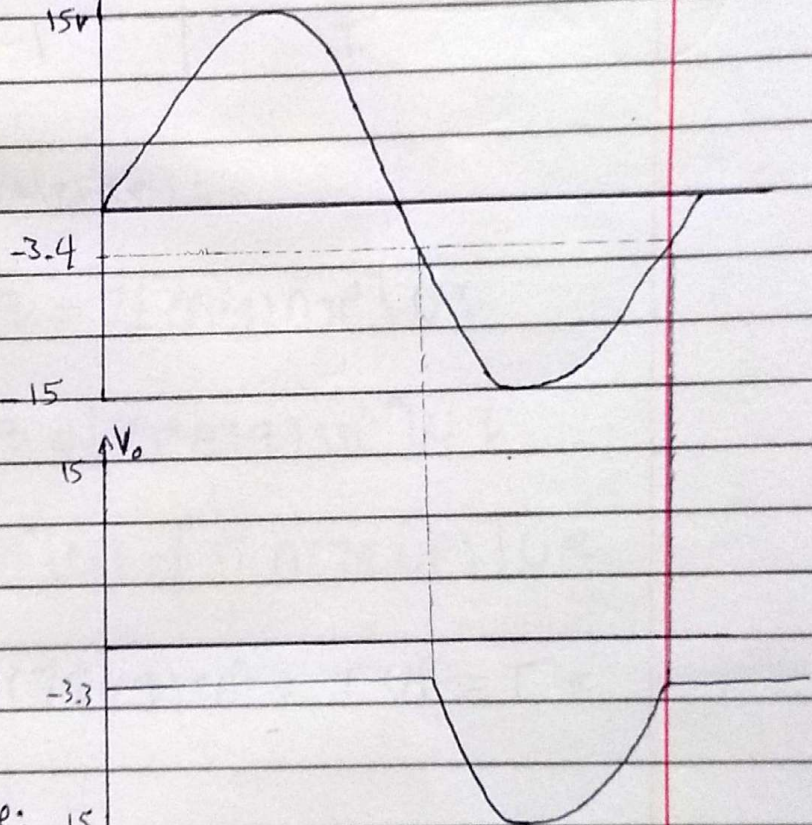
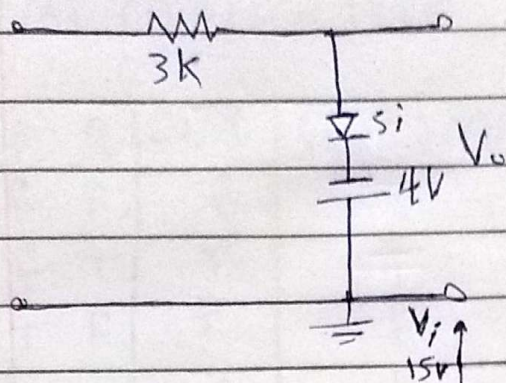
$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

$$R_B = \frac{V_{CC} - V_{BE} - I_E R_E}{I_B} = \frac{17.71 - 0.7 - 3.08 \times 0.68}{20 \times 10^{-3}}$$

$$R_B = 745.78 \text{ k}\Omega$$



Q.5) Sketch and calculate  $V_o$



\* At Positive half cycle:

Diode at on State

$$V_o = -4 + 0.7 = -3.3V$$

\* At negative half cycle:

Diode at on State when

$$V_i = 0 \text{ to } -3.3V$$

$$V_o = -4 + 0.7 = -3.3V$$

Diode at off state when

$$V_i = -3.4 \text{ to } -15V$$

$$V_o = V_i$$

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